

Systematic Volatility Trading

Financial Markets: Quant Seminar

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1. Volatility Measures

2. Gamma Trading and Replication

3. Local and Stochastic Models

4. Harvesting the VRP

Realized volatility: many definitions (and more)

Method	Data	Pseudo-formula (annualized)
Close-to-Close (CC)	Close C_t	$\hat{\sigma}_{CC}^2 = A \frac{1}{N} \sum_{t=1}^N (\ln C_t - \ln C_{t-1})^2$
Parkinson (HL)	High/Low	$\hat{\sigma}_P^2 = A \frac{1}{4 \ln 2} \frac{1}{N} \sum_{t=1}^N (\ln(H_t/L_t))^2$
Garman–Klass (OHLC)	O,H,L,C	$\hat{\sigma}_{GK}^2 \approx A \frac{1}{N} \sum \left[0.5 (\ln(H_t/L_t))^2 - (2 \ln 2 - 1) (\ln(C_t/O_t))^2 \right]$
Rogers–Satchell (OHLC)	O,H,L,C	$\hat{\sigma}_{RS}^2 = A \frac{1}{N} \sum \left[\ln(H_t/C_t) \ln(H_t/O_t) + \ln(L_t/C_t) \ln(L_t/O_t) \right]$
Yang–Zhang (OHLC)	O,H,L,C	$\hat{\sigma}_{YZ}^2 = A \left[w \hat{\sigma}_{oc}^2 + (1-w) \hat{\sigma}_{RS}^2 + \hat{\sigma}_{co}^2 \right]$
High-frequency RV	intraday r_i	$\hat{\sigma}_{HF}^2 = \sum_{i=1}^M r_i^2$ (then scale by A)
Var Swap	per term-sheet	$\hat{\sigma}_{VS}^2 = A \frac{1}{n} \sum_{t=1}^n (\ln(P_t/P_{t-1}))^2$

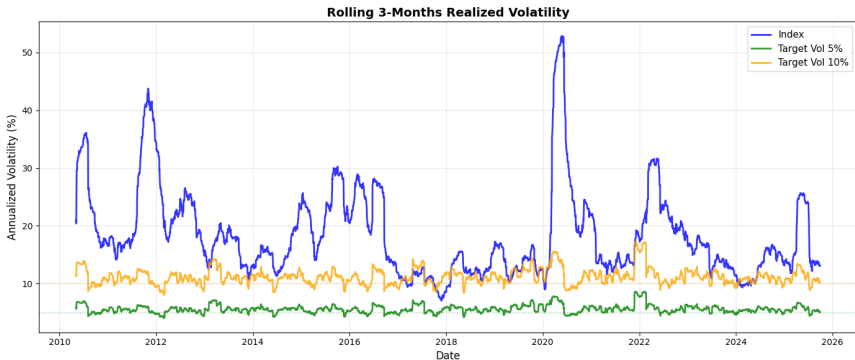
Annualization: typically $A \approx 252$ for daily data (adapt to sampling).

Volatility Targeting: what's the point

Idea: Scale exposure to maintain constant volatility over time.

$$w_t = \frac{\sigma_{\text{target}}}{\sigma_{\text{realized},t}}$$

High volatility → reduce exposure. Low volatility → increase exposure



From diffusions to quadratic variation

Under the risk-neutral measure $\mathbb{Q}$,

$$dX_t = \mu_t dt + \sigma_t dW_t, \quad X_t = \ln S_t.$$

Quadratic variation (QV) = integrated variance.

$$[X]_T = \lim_{\|\Pi\| \rightarrow 0} \sum (\Delta X)^2 = \int_0^T \sigma_t^2 dt.$$

With intraday sampling, the *realized variance* (sum of squared intraday returns) $\hat{\sigma}^2$ converges in probability to $[X]_T$.

Condition on the volatility path. Given the path $\{\sigma_t\}_{0 \leq t \leq T}$:

$$X_T \mid \{\sigma\} \sim \mathcal{N}(m_T, v_T), \quad v_T = [X]_T, \quad m_T = \int_0^T \mu_t dt - \frac{1}{2} v_T.$$

Hence $S_T \mid \{\sigma\}$ is lognormal with variance v_T .

Key identity (Black-Scholes “mixture”). Let the *root-mean-squared (RMS) volatility* be

$$\bar{\sigma}_{\text{RMS}} := \sqrt{\frac{1}{T} \int_0^T \sigma_t^2 dt} = \sqrt{[X]_T / T}.$$

Then the time-0 price of any European option satisfies

$$C_0 = \mathbb{E}^{\mathbb{Q}} \left[C^{\text{BS}}(S_0, K, T; \bar{\sigma}_{\text{RMS}}) \right]$$

Stochastic volatility, jumps, and the smile

Stochastic volatility (no jumps).

$$v_T = [X]_T = \int_0^T \sigma_t^2 dt \quad \text{is random across paths.}$$

Therefore

$$C_0 = \mathbb{E}^{\mathbb{Q}} \left[C^{\text{BS}}(S_0, K, T; \sqrt{v_T/T}) \right],$$

which is a *mixture of Black–Scholes models* (expectation of BS prices over the distribution of v_T). A mixture of vols produces a *non-flat implied-volatility smile*; its shape reflects the $\mathbb{Q}$ -distribution of v_T (more dispersion $\Rightarrow$ more curvature).

Jumps in prices.

$$[X]_T = \underbrace{\int_0^T \sigma_t^2 dt}_{\text{diffusive part}} + \underbrace{\sum_{0 < s \leq T} (\Delta X_s)^2}_{\text{jump part}}.$$

High-frequency realized variance estimates the *total* QV above. However, with jumps S_T is not conditionally lognormal, so the BS-mixture identity is an *approximation*; jumps typically add skew and fat tails (smile).

Deterministic volatility (baseline). If σ_t is deterministic, then v_T is deterministic $\Rightarrow$ a single Black–Scholes model $\Rightarrow$ *flat* smile.

Intraday realized volatility (event-based)

Key idea: Build “more i.i.d.” series with *volume/dollar bars* or *information-driven (imbalance) bars*; then compute realized variance on those bars to reduce microstructure bias and stabilize bars/day.

Observed price: $P_t^{\text{obs}} = P_t^* + \varepsilon_t$ (spread, discreteness, latency). Event bars mitigate the noise amplification of ultraHF time bars.

How to build bars.

- *Dollar/Volume bars*: print a bar after traded notional D^* or volume V^* ; adapt D^* , V^* (EWMA) to keep bars/day stable.
- *Imbalance bars (TIB/VIB/DIB)*: close when signed (notional) flow exceeds its expected run; expectations updated via EWMA.
- Optional *CUSUM* event filter to trigger only on meaningful run-ups/run-downs.

What is implied volatility and how to compute it

Definition. The *implied volatility (IV)* of a European option is the value σ_{imp} that, plugged into Black–Scholes (BS), matches the market price V^{mkt} :

$$V^{\text{BS}}(S_0, K, T; r, q; \sigma_{\text{imp}}) = V^{\text{mkt}}.$$

It is a *risk-neutral* parameter summarizing the price of diffusion risk (plus risk premia embedded in option prices). **Computation (numerical inversion).**

- Root-finding on $f(\sigma) = V^{\text{BS}}(\sigma) - V^{\text{mkt}}$ (e.g., Newton–Raphson, Brent).
- Good practice: use *vega*-aware updates; cap σ and guard for low vega (deep ITM/OTM, very short maturities).
- For puts: same inversion (or convert via put–call parity).

Surface. Repeat across (K, T) to obtain the *IV surface* $\sigma_{\text{imp}}(K, T)$.

From points to a surface: construction & no-arbitrage

Pre-processing.

- Infer forward $F(T)$ and discounting from put–call parity (and dividends).
- Clean quotes (crossed markets, stale prints), unify conventions (BP vol vs decimal).

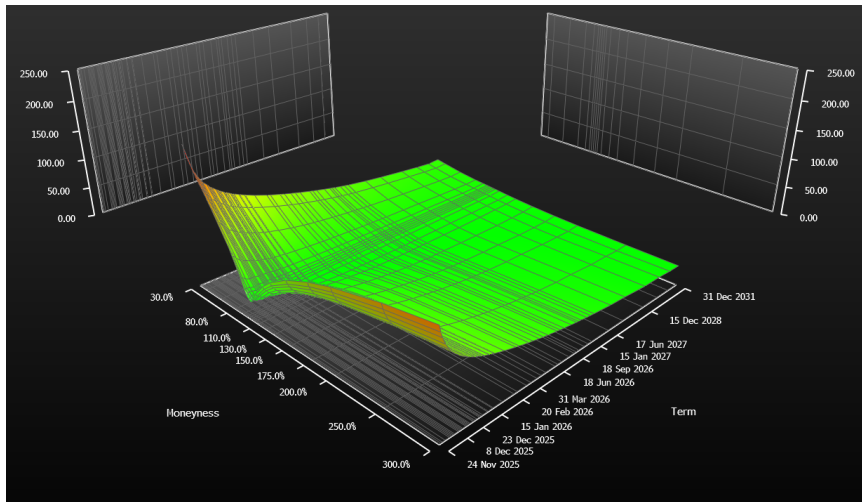
Fitting / interpolation.

- Parameterizations (e.g., SVI/SSVI, polynomial-in-moneyness, mixture-lognormal) or local smoothing in $(\log K, \sqrt{T})$.
- Interpolate in a monotone/convexity-preserving space; extrapolate wings with asymptotics consistent with tails.

Static no-arbitrage constraints.

- *Vertical (butterfly) arbitrage*: call price convex in strike $\Rightarrow$ non-negative densities.
- *Calendar arbitrage*: total variance $w = \sigma^2 T$ non-decreasing in T .
- *Butterfly & calendar* must hold jointly after smoothing; check also put–call parity.

Volatility Surface



Main inconsistencies & practical pitfalls

Model inconsistency (BS vs reality)

- BS assumes *single* σ , yet markets display **smile/skew** and **term structure** $\Rightarrow$ IV becomes a surface.
- Under jumps/stochastic vol, IV is not a forecast of future σ : it reflects risk-neutral averages and risk premia, not physical expectations.

Surface-level inconsistencies

- Violations of *static no-arbitrage*: non-convex slices (butterfly arb), decreasing total variance in T (calendar arb).
- Wings extrapolated too flat/steep $\Rightarrow$ negative densities or unrealistic tail probabilities.

Operational pitfalls

- Low vega regions (deep ITM/OTM, short T) $\Rightarrow$ noisy IV.
- Dividend/forward mis-estimation contaminates the entire surface.
- Mixing American/European conventions leads to biased IVs.

Variance Risk Premium

Definition. Over a horizon T , the *variance risk premium* is the gap between the *risk-neutral* expectation of integrated variance and its *physical* expectation:

$$\text{VRP}(T) := \underbrace{\mathbb{E}^{\mathbb{Q}} \left[\frac{1}{T} \int_0^T \sigma_t^2 dt \right]}_{\text{implied (variance swap fair strike)}} - \underbrace{\mathbb{E}^{\mathbb{P}} \left[\frac{1}{T} \int_0^T \sigma_t^2 dt \right]}_{\text{expected realized}}$$

Empirically, one often proxies $\text{VRP}(T)$ by $\mathbf{IV}^2(T) - \mathbf{RV}^2(T)$ at matching maturities. On equity *indexes*, it tends to be positive (insurance/hedging demand, jump risk, hedging frictions).

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1) Setup and Black–Scholes PDE

Market setup. Let the underlying follow under P :

$$dS_t = \mu S_t dt + \sigma_R(t) S_t dW_t,$$

with risk-free rate r (no dividends; extendable to a continuous yield q).

Option priced at implied vol. Consider $V(S, t; \sigma_I)$, a European

option valued by BSM with *implied* volatility σ_I (treated as constant on $[t_0, T]$). Define Greeks at σ_I :

$$\Delta_I = \partial_S V, \quad \Gamma_I = \partial_{SS} V, \quad \Theta_I = \partial_t V.$$

BSM PDE at σ_I :

$$\boxed{\Theta_I + \frac{1}{2} \sigma_I^2 S^2 \Gamma_I + rS\Delta_I - rV = 0.} \quad (1)$$

(With dividends q : replace $rS\Delta_I \rightarrow (r - q)S\Delta_I$.)

2) Itô dynamics along realized-vol paths

Apply Itô to $V(S_t, t; \sigma_I)$ when S_t evolves with *realized* volatility $\sigma_R(t)$:

$$\begin{aligned} dV_t &= \Theta_I dt + \Delta_I dS_t + \frac{1}{2} \Gamma_I (dS_t)^2 \\ &= \Theta_I dt + \Delta_I (\mu S_t dt + \sigma_R(t) S_t dW_t) + \frac{1}{2} \Gamma_I \sigma_R^2(t) S_t^2 dt. \end{aligned} \quad (2)$$

The pricing model remains BSM at σ_I (PDE (1)), while the *path* of S_t is driven by $\sigma_R(t)$.

3) Delta-hedged portfolio and self-financing condition

Portfolio: long option, short Δ_I shares:

$$\Pi_t := V_t - \Delta_I S_t.$$

Self-financing P&L (continuous trading):

$$d\Pi_t = dV_t - \Delta_I dS_t - r(V_t - \Delta_I S_t) dt. \quad (3)$$

Insert (2) into (3) and cancel dS_t terms:

$$d\Pi_t = \left(\Theta_I + \frac{1}{2} \Gamma_I \sigma_R^2 S_t^2 - r(V_t - \Delta_I S_t) \right) dt. \quad (4)$$

Use the PDE (1) to replace $\Theta_I - r(V_t - \Delta_I S_t)$:

$$\Theta_I - r(V_t - \Delta_I S_t) = -\frac{1}{2} \sigma_I^2 S_t^2 \Gamma_I.$$

4) Central identity and present value of P&L

Central identity (gamma trading):

$$d\Pi_t = \frac{1}{2} \Gamma_I(t) S_t^2 (\sigma_R^2(t) - \sigma_I^2) dt. \quad (5)$$

Present value (continuous compounding):

$$PV[\text{P\&L}] = \frac{1}{2} \int_{t_0}^T e^{-r(t-t_0)} \Gamma_I(t) S_t^2 (\sigma_R^2(t) - \sigma_I^2) dt. \quad (6)$$

Interpretation: the P&L is the time integral of the *gamma-dollar* $\Gamma_I S^2$ times the variance gap (realized minus implied), discounted at r .

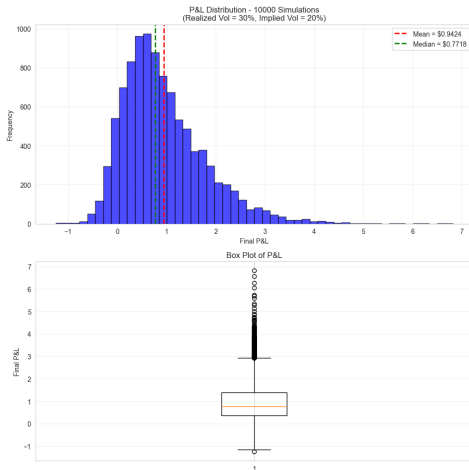
Alternative hedging schemes

- **Hedge at realized vol (Δ_R):** If one could delta-hedge using the true (unknown ex-ante) σ_R , the discounted P&L becomes *deterministic* and equals the difference of BSM prices evaluated at σ_R vs Σ .
- **Hedge at implied vol (Δ_I):** Realistic case. The P&L follows (5)–(6); it is path-dependent because $\Gamma_I(t)S_t^2$ depends on S_t .
- **Discrete rebalancing and frictions:** Finer re-hedging reduces variance of P&L but costs rise; jumps and vol-of-vol add correction terms beyond (5).

Trading gamma vs quadratic variation

- A long-gamma, delta-hedged position *harvests convexity*: it benefits when the *quadratic variation* of S exceeds what the model (via σ_I) priced in.
- The sign of expected P&L is driven by $\sigma_R^2 - \sigma_I^2$; the magnitude is weighted by $\Gamma_I S^2$, largest near-the-money and for shorter maturities.
- Forward vs spot as hedging instrument (with dividends/rates) affects implementation but not the core identity.

P&L vs Volatility Difference



Key Insight:

P&L is driven by volatility:

$$P\&L \propto (\sigma_{\text{real}} - \sigma_{\text{impl}})$$

Trading Implications:

- Long options (long gamma) when you expect realized vol $>$ implied vol
- Short options (short gamma) when you expect realized vol $<$ implied vol

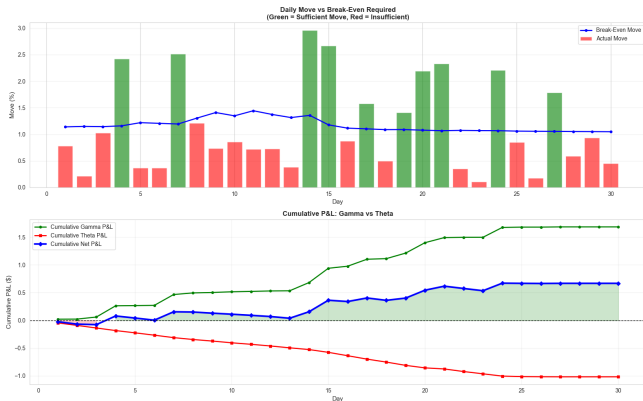
Gamma vs Theta Trade-off

Break-Even Formula:

$$\Delta S_{BE} = \sqrt{\frac{2|\Theta|}{\Gamma}}$$

Key Insights:

- Short-dated: larger % moves
- BE move $\approx$ daily implied vol



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Motivation: Why (and when) go beyond Black–Scholes?

Limitations of constant-vol BSM:

- Cannot reproduce the entire implied volatility surface (smile/skew & term structure).
- Lacks *dynamics* of the surface (how smile moves through time/moneyness).
- Path-dependence matters: barriers, cliquets, forward-starts are sensitive to the *forward* smile.

Two families that address these issues:

- 1 **Local Volatility (LV):** $\sigma_{\text{loc}} = \sigma_{\text{loc}}(S, t)$ deterministic; *exact fit* to today's vanilla surface.
- 2 **Stochastic Volatility (SV):** volatility is a random process (vol-of-vol, correlation ρ); better *dynamic* realism.

Local Volatility (LV): model, calibration, properties

Dynamics (under Q):

$$dS_t = (r - q)S_t dt + \sigma_{\text{loc}}(S_t, t) S_t dW_t.$$

Dupire formula (call forward-price $C(K, T)$):

$$\sigma_{\text{loc}}^2(K, T) = \frac{2 \partial_T C(K, T)}{K^2 \partial_{KK} C(K, T)} \quad (\text{numerically}).$$

Pros: exact fit to today's vanillas; stable numerics; arbitrage control across K, T (with regularization).

Cons: no vol-of-vol; “sticky-strike” dynamics often too rigid for equity-type skews; forward smile may be unrealistic for some exotics.

Stochastic Volatility (SV): model, dynamics, calibration

Generic SV (e.g., Heston-like) under $\mathbb{Q}$:

$$dS_t = (r - q)S_t dt + \sqrt{v_t} S_t dW_t^{(1)},$$
$$dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^{(2)}, \quad d\langle W^{(1)}, W^{(2)} \rangle_t = \rho dt.$$

Key features:

- *Vol-of-vol* ξ and mean reversion κ shape curvature and term structure of smile.
- *Correlation* $\rho < 0$ (equity) generates negative skew.
- Implied surface *dynamics* more realistic than LV.

Pricing & calibration:

- Semi-closed forms (characteristic functions) or numerical.
- Fit ATM term + major slices; exact global fit not guaranteed.

Practical guidance and extensions

When to prefer LV:

- You need an exact fit to today's vanilla surface and a fast/stable engine for many path-dependent but “smile-insensitive” payoffs.

When to prefer SV:

- Products sensitive to *forward* smile (barriers, forward-starts, cliquets); need realistic skew dynamics (e.g., $\rho < 0$), term structure, and convexity from vol-of-vol.

Hybrid approach (SLV):

- *Stochastic Local Volatility*:

$dS_t = (r - q)S_t dt + \sigma_{\text{loc}}(S_t, t)\sqrt{v_t} S_t dW_t$ combines exact vanilla fit (via σ_{loc}) with realistic dynamics (via v_t).

Implementation checklist:

- 1 Clean & arbitrage-free surface (smoothing).
- 2 Choose LV or SV (or SLV) by product sensitivity to forward smile.
- 3 Calibrate (global + dynamic targets), validate on exotics, backtest hedges.

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Letting volatility move: live σ_t and the three extra terms

Itô with fixed σ :

$$dV_t = \Theta_I dt + \Delta_I dS_t + \frac{1}{2} \Gamma_I d\langle S \rangle_t \Rightarrow \boxed{d\Pi_t = \frac{1}{2} \Gamma_I \left(d\langle S \rangle_t - \sigma^2 S_t^2 dt \right)}.$$

Now let the live implied volatility float: consider $V(S, t; \sigma_t)$ and apply Itô in two variables:

$$dV_t = \Theta_I dt + \Delta_I dS_t + \frac{1}{2} \Gamma_I d\langle S \rangle_t + \underbrace{V_\sigma d\sigma_t}_{\text{vega}} + \underbrace{\frac{1}{2} V_{\sigma\sigma} d\langle \sigma \rangle_t}_{\text{volga}} + \underbrace{V_{S\sigma} d\langle S, \sigma \rangle_t}_{\text{vanna}}.$$

With the same portfolio and the PDE at σ_t :

$$\boxed{d\Pi_t = \frac{1}{2} \Gamma_I \left(d\langle S \rangle_t - \sigma_t^2 S_t^2 dt \right) + V_\sigma d\sigma_t + \frac{1}{2} V_{\sigma\sigma} d\langle \sigma \rangle_t + V_{S\sigma} d\langle S, \sigma \rangle_t.}$$

The vega-gamma identity under BSM

For a call:

$$V_{\sigma} = S e^{-q\tau} \phi(d_1) \sqrt{\tau}, \quad \Gamma_I = \frac{e^{-q\tau} \phi(d_1)}{S \sigma \sqrt{\tau}} \Rightarrow \sigma \tau \Gamma_{\$} = V_{\sigma}.$$

Dollar-gamma: $\Gamma_{\$}(t) := S_t^2 \Gamma_I(t)$,

time-to-maturity: $\tau_t := T - t$.

Key identity:

$$V_{\sigma} = \sigma_t \tau_t \Gamma_{\$}(t).$$

It links the vega exposure to the gamma-dollar and will be crucial to disentangle the vega block in the P&L from the pure gamma block.

Disentangling step I: rewrite the vega block

Use Itô on σ_t^2 : $d(\sigma_t^2) = 2\sigma_t d\sigma_t + d\langle\sigma\rangle_t$.

Combine the three “vol” pieces in the PDE using $V_\sigma = \sigma_t \tau_t \Gamma_\(t) :

$$V_\sigma d\sigma_t + \frac{1}{2} V_{\sigma\sigma} d\langle\sigma\rangle_t + V_{S\sigma} d\langle S, \sigma\rangle_t$$

$\implies$ (total Itô derivative) + terms in $(\sigma_t - \sigma_0) d\Gamma_t^*$,
where $\sigma_0 := \sigma_{t_0}$ and $\Gamma_t^* := e^{-rt} \Gamma_\(t) (discounted dollar-gamma).

Intuition: one piece is a pure differential that is 0 at t_0 and $T \Rightarrow$ its *integral over life is zero* (the “vega term”). The other piece is the small “dGamma term”.

Disentangling step II: the final pathwise identity

Putting everything together yields the pathwise P&L increment (with traded vol σ_0 in the gamma block):

$$\begin{aligned} d\Pi_t = & \frac{1}{2} \Gamma_I \left(d\langle S \rangle_t - \sigma_0^2 S_t^2 dt \right) \\ & + d[\text{vega-term}] + (\sigma_t - \sigma_0) d\Gamma_t^* + \text{residual drifts.} \end{aligned}$$

Consequences:

- The **vega-term** integrates to zero over $[t_0, T]$ (adds MTM noise only).
- The **dGamma term** is typically small (in BSM, Γ^* is a martingale $\Rightarrow$ zero mean).
- The **dominant driver** of cumulative performance is the *pure gamma* term at traded volatility σ_0 .

Reading the formula over the life of the trade

Over $[t_0, T]$:

$$\int_{t_0}^T d[\text{vega-term}] = 0$$

$$\Rightarrow \text{Cumulative P\&L} \approx \frac{1}{2} \int \Gamma_I (d\langle S \rangle - \sigma_0^2 S^2 dt) + \int (\sigma_t - \sigma_0) d\Gamma_t^*.$$

Takeaways.

- The main contribution behaves like a **realised-minus-traded variance** weighted by $\Gamma_{\S}$.
- The dGamma correction averages out in many markets; what remains is the “variance gap” intuition.

Use: Clean attribution, the sensitivity to the volatility premium is the *pathwise average* of dollar-gamma scaled by tenor and by $(\sigma_r + \sigma_0)/2$.

Empirical Illustration: Selling Straddles

Setup:

- We consider a systematic short-gamma strategy on the **EURO STOXX 50 (SX5E)**.
- At each rebalancing date, we:
 - sell a 1-month **ATM straddle** at traded implied volatility σ_0 ,
 - delta-hedge the position daily using futures,
 - roll the position each month at fixed maturity.
- We analyse the cumulative P&L via the pathwise identity:

$$d\Pi_t = \text{pure gamma} + \text{vega-term} + d\text{Gamma-term} + \text{residuals}.$$

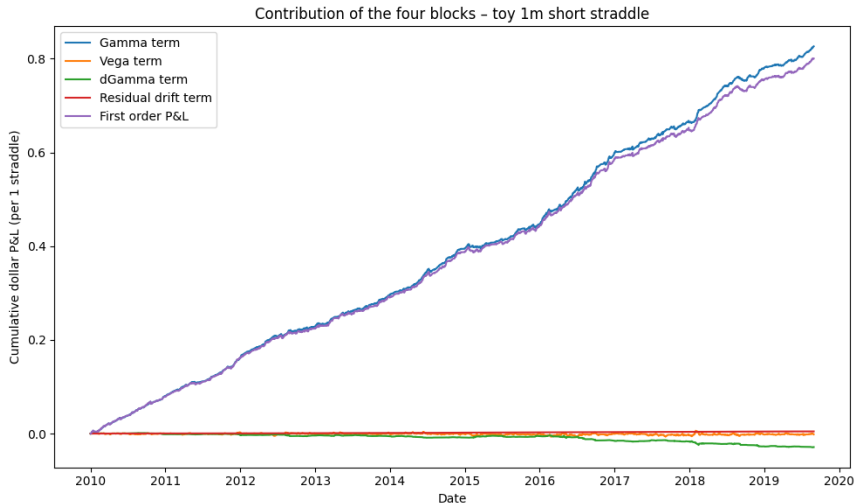
Delta Hedged Portfolio: Historical Backtest



Straddle Pathwise Identity

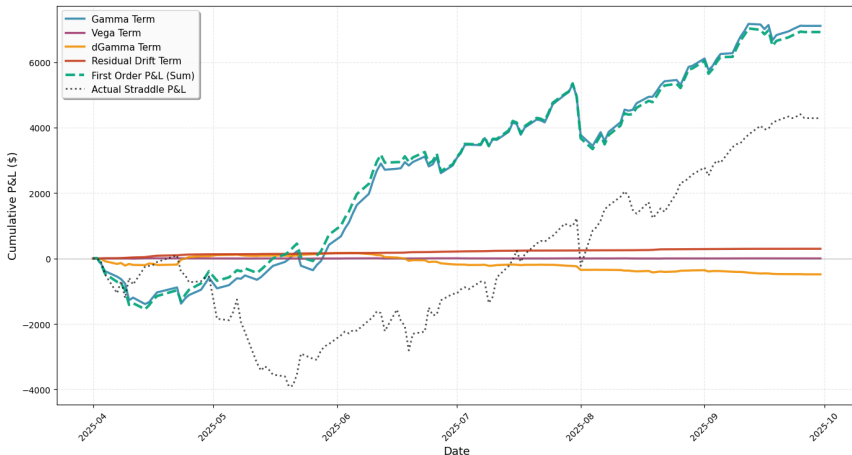
$$\begin{aligned}
 \text{P\&L} = & \left[\underbrace{t \bar{\Gamma}^* \frac{\sigma_r + \sigma_0}{2} (\sigma_r - \sigma_0)}_{\text{Volatility premium component}} + \underbrace{\frac{t}{2} \rho \text{Stdev}(\Gamma^*) \text{Stdev}\left(\frac{d\langle S \rangle}{S^2 du}\right)}_{\text{Gamma covariance effect}} \right. \\
 & + \underbrace{e^{-rt} \frac{\sigma + \sigma_0}{2\sigma} \frac{\partial P}{\partial \sigma}(t) (\sigma - \sigma_0)}_{\text{Vega term}} \\
 & - \underbrace{\int_0^t \frac{T-u}{2} (\sigma^2 - \sigma_0^2) d\Gamma_u^*}_{\text{dGamma term}} \\
 & \left. - \underbrace{\int_0^t \frac{e^{-ru}}{2} \left(\frac{\nu}{\sigma} \frac{\partial \nu}{\partial \sigma} \right) d\langle \sigma \rangle_u}_{\text{Residual drift term}} \right]
 \end{aligned}$$

Toy Model: Positive VRP



Trump and Liberation Day: Goodbye, Log-Normality

P&L Decomposition during Liberation Day



Conclusions

- Empirical evidence from econometric studies shows that most assets deviate significantly from the log-normal assumption.
- Traders rely on the Black framework as a quoting tool, yet routinely operate outside its core assumptions—an inconsistency that resurfaces when computing sensitivities.
- To mitigate P&L instability, traders must adopt models that better capture the true dynamics of the underlying. This has led to the widespread use of local volatility, stochastic volatility and jump models.
- Statistical analysis and trading practice ultimately reveal the same issue: inadequate modeling of the underlying. What econometrics detects as model misspecification manifests as hedging difficulties for traders.

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